Summer Term 2026

Investment and Financial Management
Exercise on Mathematical Basics

Session 06: Bonds I

Center for Digital Transformation
Technische Universität München (TUM), Campus Heilbronn

Government bonds

The U.S Treasury is launching – or rather re-launching – the 20-year bond, as a way to fund its ongoing budget deficits. The 20-year hadn't been sold to the public regularly since 1986.

On Wednesday, the Treasury provided final details on its plans, saying it will offer $20 billion in 20-year bonds in an initial auction scheduled for May 20.

"Treasury is rebooting issuance of the 20-year bond as it seeks to lock in historically low interest rates to help contain taxpayer cost on its growing pile of debt," says Daniel Milan, managing partner of Cornerstone Financial Services in Southfield, Michigan.

Source: Bankrate, May 6, 2020

Bonds: Introduction

Which statement is correct about bonds?

1 2

(A) Whereas some types of bonds are debt instruments, others can also be described as hybrid instrument – a combination of securitized debt and equity claims.

(B) Compared to the overall market value of stocks, bonds do not seem to play a major role in financial markets.

(C) Due to the fact that future cash flows of bonds are in general very hard to predict, e.g. compared to stocks, applying the basic pricing principle is much more complicated.

(D) While the creditworthiness of the issuer counts as one of the major factors influencing the bond price for private corporations, this does not hold for countries or states when issuing bonds.

Bonds: Introduction

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1

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Bonds are long-term contracts with issuers (borrowers) committing themselves to repay debt at a specified date and, if applicable, to pay coupon payments at predetermined points in time to the bond owner

- Since a bond is a debt instrument, usually its cash flows can be easily predicted.
  - Exception: variable coupon
  - Insolvency risk (credit risk)
- The discount or interest rate applied in the pricing formula equals the return of an alternative investment with a similar level of risk.

Pricing Principle

As real investments, financial investments can also be priced using the ***Discounted Cash Flow-Principle***.

Introduction – Definition and Pricing Principle

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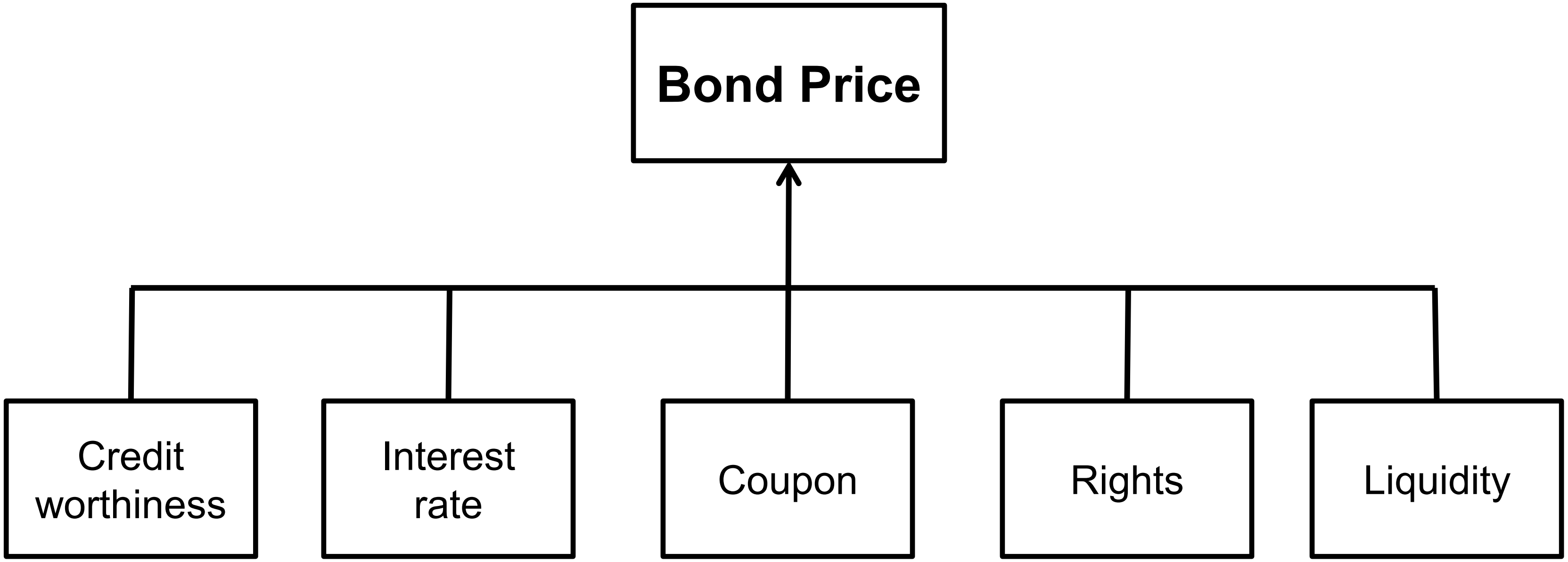


Introduction – Definition and Pricing Principle

Bond	Feature
Coupon Bond	Bond with fixed interest payments over lifetime (coupon payments) and repayment of principle amount at maturity of the bond
Zero-coupon bond (zerobond)	Bond without coupon payments with repayment of face value at maturity
Bond with variable interest (Floater)	Coupon payments are adjusted to current interest level (LIBOR)
Perpetual / Consol Bond	Bonds without maturity date
Convertible Bond	Bond with small coupon that provide the possibility to be exchanged against stocks at a predetermined date in time
Option bond / warrant bond	Follows the idea of a convertible bond, although the option right is traded separately
Reverse convertibles	The issuer can choose to repay with shares instead of cash

Introduction – The market for stocks and bonds

Market (in billion US$) End 2011	Stocks (1)		Bonds (2)	
U.S.	15.640,71	32,96%	33.693,90	34,37%
Europe (ex UK)	6.190,0	13,05%	30.404,50	31,02%
China (incl HK)	5.670,1	11,95%	3.691,90	3,77%
Japan	3.540,7	7,46%	15.275,00	15,58%
UK	3.266,4	6,88%	4.813,80	4,91%
India	1.992,5	4,20%	653,80	0,67%
Canada	1.912,1	4,03%	2.246,00	2,29%
Brazil	1.228,9	2,59%	1.697,10	1,73%
Australia	1.198,2	2,53%	1.683,80	1,72%
Switzerland	1.089,5	2,30%	769,70	0,79%
South Korea	996,1	2,10%	1.312,90	1,34%
South Africa	789,0	1,66%	246,70	0,25%
Russia	783,6	1,65%	164,90	0,17%
Taiwan	635,5	1,34%		
Singapore	598,3	1,26%	193,30	0,20%
Mexico	408,7	0,86%	567,90	0,58%
Malaysia	395,6	0,83%	41,80	0,04%
Other	1.111,34	2,34%	573,40	0,58%
Total	47.447,3	100.0%	98.030,4	100,0%
% of total stocks & bonds		32,61%		67,39%
Total stock & bond markets	145.477,7			

12/2018:

Stocks: 68,654 (38.4%)

Bonds: 110,000 (61.6%)

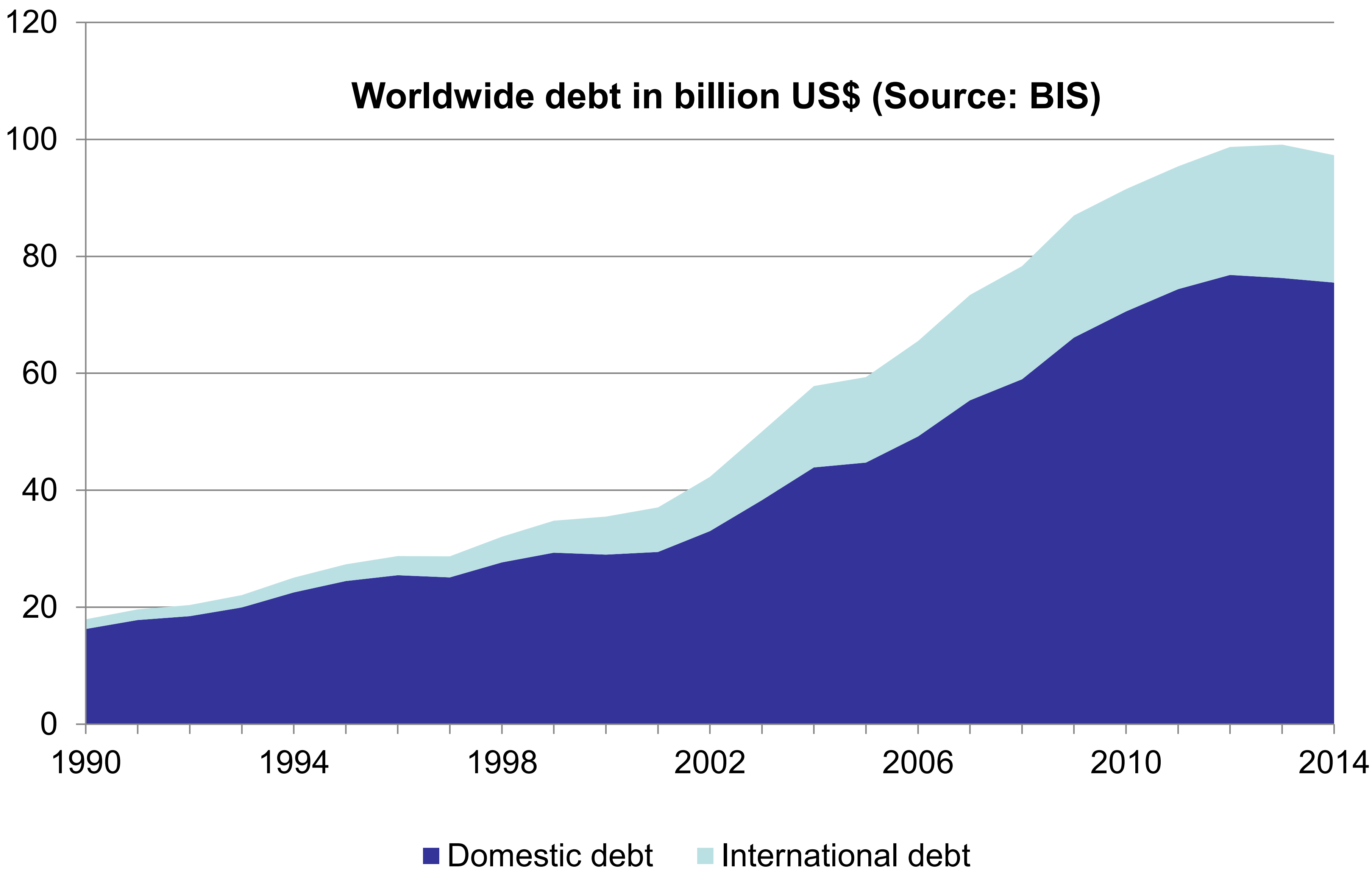
World-GDP: 84,800

Source: World Federation of Exchanges (as of 31-Dec-2011) for Stock Market Data, Bank for International Settlements (as of 31-Dec-2011) for Bond Market Data

(1) stock market values are based on market capitalization (market values)

(2) bond market values are based on debt outstanding (book values) and contain domestic and international debt (to foreigners).

Introduction – The market for stocks and bonds



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Bond Pricing – Terms and Symbols

Bond	Feature
B_0	$Price\ of\ a\ coupon\ bond\ (in\ percent\ of\ nominal\ /\ face\ value)$
B_0^{ZB}	$Price\ of\ a\ zerobond\ (in\ percent\ of nominal\ /\ face\ value)$
B_N	$Redemption\ value\ (in\ percent\ of\ nominal\ /\ face\ value)$
C	$Coupon\ (in\ percent\ of\ nominal\ /\ face\ value)$
C_k	$Coupon\ payment\ at\ time\ k$
r	$risk-adjusted\ discount\ or\ interest\ rate$
m	$number\ of\ coupon\ payments\ per\ year$
N	$Maturity\ of\ the\ bond\ in\ number\ of\ periods$
t_1, t_0	$Time$
k	$Time\ index$
D	$Duration\ of\ the\ bond$
D_{mod}	$modified\ duration\ of\ the\ bond$

Bond Pricing

The pricing formula for bonds can be derived from the established present value formulas for the evaluation of investment projects because it follows the same logical idea

General present value formula

$$(1): PV = \sum_{n=0}^{N} \frac{CF_n}{(1+r)^n}$$

Equivalent Principle

The present value of a group of cash flows needs to be sum of the all single present values of the cash flows (additive linking).

Underlying assumption at financial markets

Rational market participants price interest instruments depending only on the cash flows that are directly related to the underlying instrument.

Pricing of Zerobonds

A zerobond (zero coupon-bond) is a fixed-income security that is repaid after N years with no intermediate coupon payments

Interest claims are not cleared every period but are rather considered in the reduced price of the security compared to its nominal (face) value. **That is why zerobonds are called discount security.**

The **price of a zerobond equals the redemption amount discounted with the risk-adjusted interest rate to the date of issuance**. The formula equals the valuation of an annuity-immediate with a one-time payment at time N.

$$(F): B_0^{ZB} = \frac{B_N}{(1+r)^N} = B_N * q^{-N}$$

0	1	2	…	t-1	t	…	N-1	**N**
-	-	-	-	-	-	-	-	$\boldsymbol{B_N}$

Pricing of Zerobonds

(A.1a) The effective risk-adjusted interest rate should equal 8%. The Bamboo Cars AG issues zerobonds with a maturity of 25 years which are repaid at a face value of 100 EUR. Please compute the issue price for today.

0	1	2	…	t-1	t	…	N-1	**25**
-	-	-	-	-	-	-	-	$\boldsymbol{B_N = 100}$

$\boldsymbol{B_0^{ZB}} = ?$

$$(1): B_0^{ZB} = \frac{B_N}{(1+r)^N} = \frac{100}{1.08^{25}} = 14.60\ EUR$$

An investor who buys the bond at issuance and holds it until maturity will earn the **risk-adjusted market interest** rate of 8.00%.

$$(2)\ r = \sqrt[25]{\frac{100}{14.60}EUR} - 1 = 8.00\%$$

Pricing of Zerobonds

(A.1b) The effective risk-adjusted interest rate should equal 8%. The Bamboo Cars AG issues zerobonds with a maturity of 25 years which are repaid at a face value of 100 EUR. Please compute the issue price for today.

After ten years the effective annual interest rate has dropped to 6%. What is the market value of the bond?

0	1	2	…	10	11	…	24	**25**
-	-	-	-	-	-	-	-	$\boldsymbol{B_N = 100}$

$$\boldsymbol{B_{10}^{ZB}} = ?$$

$$(1): Years\ to\ maturity = 15\ \text{years}; \mathrm{r_{new}} = 6\%$$

$$(2): B_{10}^{ZB}(6\%) = \frac{100}{1.06^{15}} = 41.73\ EUR$$

$$(3): B_{10}^{ZB}(8\%) = \frac{100}{1.08^{15}} = 31.52\ EUR$$

Pricing of Zerobonds

(A.1c) The effective risk-adjusted interest rate should equal 8%. The Bamboo Cars AG issues zerobonds with a maturity of 25 years which are repaid at a face value of 100 EUR. Please compute the issue price for today.

Which effective annual interest does an investor earn if the investor buys the security at issuance and sells the bond at $k = 10$?

$$(1): B_N = B_0 * q^N$$

$$(2): q = \sqrt[N]{\frac{B_N}{B_0}} = \sqrt[10]{\frac{B_{10}}{B_0}} = \sqrt[10]{\frac{41.73}{14.60}} = 1.1107 \Rightarrow r = 11.07\%$$

Gain due to drop of interest level!

0		10	25
$B_0^{ZB} = f(r = 8\%)$	$\xrightarrow{r = 11.07\%}$	$B_{10}^{ZB} = f(r = 6\%)$	$B_N = 100$

Bond becomes more attractive due to drop in interest level

Pricing of Zerobonds

(A.2a) A zerobond with a face value of 100 is issued at a real annual interest rate of 6.75%. The bond has a maturity of 20 years.

Please compute the issue price.

1 2

$A = 15.80\ EUR$	$B = 27.08\ EUR$	$C = 106.75\ EUR$	$D = 100.00\ EUR$

$$(1): B_0^{ZB} = \frac{B_N}{(1+r)^N} = \frac{100}{1.0675^{20}} = 27.08\ EUR$$

1 2

(A.2b) After 8 years the real annual interest rate changes to 6.25%. Which statement is correct without limitations?

A: An investors who has bought the bond at issuance and sells it now will earn an annual return of 7.50%.	B: This drop in the interest rate will affect the annual return of the investor, in particular if the investor holds the bond until maturity.

$$(2): B_8^{ZB} = \frac{100}{(1.0625)^{12}} = 48.31\ EUR \Rightarrow r^* = \sqrt[8]{\frac{48.31}{27.08}} - 1 = 7.50\%$$